

Biological Evaluation of Black Box Models vs. SHAP

Framework and Impact

Artificial Intelligence (AI) has been increasingly applied to solve complex real-world problems, but where we can produce promising results, explanations are less so simple. In most research fields, machine learning (ML) models like neural networks have caused former methods of analyzing data to fade in popularity in favor of ML's superior ability to uncover complex relationships between inputs and outputs in large-scale datasets.

This new shift of method brings forth new complications and in turn, necessary precautions. Whereas the use of ML makes it easier to discern information from datasets, it lacks the ability to provide an explainable result comprehensible to people. Their "black box" nature is particularly an issue for biological research used in health applications where validation and trust of information is required. To combat this, recent advancements in AI research seeks to decipher the model's internal computations by tracing their decision-making processes to understand the key-features driving its predictions.

In particular, post-hoc methods such as certain explainable AI (xAI) models can be used to extract pivotal features from datasets in an attempt to elude a ML model's black box nature. However, model-interpretation and model-explanation algorithms, like xAI, were developed for image processing and other domains, and remain largely untested with biological data. There are also some common concerns of how xAI methods will handle high-dimensional datasets prone to the "curse of dimensionality," where algorithms struggle with high-dimensional data as the data becomes sparse, requiring exponentially more samples to maintain density and leading to a higher risk of overfitting (where a model learns information strictly applicable to the training data, leading to poor performance in real-world applications) (Oldenburg, Jan, et al. 2024).

This can cause unstable interpretability in the extraction of crucial features, resulting in loss of information when extracted and or a misrepresentation of our results. An issue like this would likely be detrimental to research areas using biological data, especially in medical research, where trust and validation of information as well as replicable results are needed. This

is further complicated by the already complex nature of biological data, adding another layer of “dimensionality” to comprehend.

Thus our research intends to test the effectiveness of the xAI method SHapley Additive exPlanations (SHAP) on biological data, in which in our case, is a compendia of approximately 20,000 samples of *Escherichia coli* (*E. coli*) collected from pre-existing publicly available data deposited up until the year 2025. SHAP analysis is one of the more commonly used xAI methods, and provides a view into the inner workings of black-box ML models through calculating the Shapley values (SHAP values) of each feature individually.

SHAP values are numbers representing the contribution of each feature making up the results, giving us an understanding of what key features are contributing to the reconstruction of our input data. Essentially, SHAP values can be understood as the weighted average of a feature’s contribution to all possible outcomes (Ponce-Bobadilla, Ana Victoria, et al. 2024).

Goals

As SHAP is one of the more preferred xAI methods in most other fields of research due to its superiority in offering detailed, consistent, and actionable insights in their outputs, if we can answer what SHAP’s ability to comprehend black box models of biological datasets is, it could help to solve the issue of verification when using black box ML models to process biological data, and possibly allow for better explanation of biological information to be discerned from the dataset’s SHAP values as opposed to what would be extracted from our black box model.

Our aim is to answer if there’s information to be gained from the SHAP values of our black box that would otherwise not be present. In relation to SHAP’s success when applied to datasets of other scientific fields, such as civil engineering and demography, we anticipate SHAP to show similar success when applied to biology.

Plans

To test our hypothesis, we’ll build ADAGE models (a denoising autoencoder for gene expression) to use as our black box model, and we’ll follow a similar process to most other machine learning workflow consisting of three phases: data preprocessing, model training, and

model evaluation. For our data preprocessing phase, our *E. coli* datasets are largely prepared and require minimal work.

To start training our model, there are necessary choices to be made in our model's hyperparameters. Hyperparameters are a configuration variable set before training a model. They define the model's characteristics, dictating the model's structure and the algorithm's learning process. They affect not only the model's performance and resource requirements, but also the biology the model can pick up. Traditionally, hyperparameters are set manually through a process of trial and error, but progress has been made to automate the process, such as AutoML frameworks, to find the hyperparameters that allow for the best performance in the model – in terms of loss minimization – in a more efficient manner (Khan, Muhammad Salman, et al. 2025).

We plan to use KerasTuner to find the hyperparameters of our model. Tuners, like KerasTuner, choose the parameter combinations based on minimizing loss, however, that does not necessarily mean they pick the parameters that allow for the best biology. Thus after the assessment framework of our model has been set up, there is a possibility of continuing our research by testing the optimal hyperparameters that may allow for more biology to be picked up when using SHAP by incorporating biological evaluations into our tuner's framework.

But for now, a few hyperparameters to be considered for our model is the size of the input data, including what portion of the dataset to allocate for training versus the portion to be reserved for testing our model later. Later, when employing SHAP, we'll also have to decide how to go about masking our data for our model and the manner in which we weigh our SHAP values. Once our hyperparameters are established, we can go about training our model. The training phase in its simplest form is a process of multiplying varying weights to the input vector until it has constructed a model with the least amount of error between the input and the reconstructed output. Once our ADAGE models are constructed and frozen, we can use Python's SHAP package to calculate our SHAP value of each sample for each feature.

For our model evaluation phase, we're testing our models performance in biological evaluation and comparing their results. To do so, we're using the models' weight matrices to compare the gene-gene correlations inside to our pre-defined biologically annotated database, EcoCyc, that shows how the genes should be related to each other. The model with more congruence to the annotated database is the model that performs better in extracting biological information.

Personal Goals

My main priority for this project is to practice using machine learning models effectively, particularly when handling biological data. I plan to pursue a career in computational biology, but I lack relevant experience in regards to the practical applications of the field and would like to start building the necessary frameworks. This opportunity would allow me the chance to work towards my goal in a familiar environment with like-minded people.

Works Cited

- Khan, Muhammad Salman, et al. "Comparative Analysis of Automated Machine Learning for Hyperparameter Optimization and Explainable Artificial Intelligence Models." *IEEE Access*, vol. 13, 2 May 2025, pp. 84966–84991, <https://doi.org/10.1109/access.2025.3566427>.
- Oldenburg, Jan, et al. "XModNN: Explainable Modular Neural Network to Identify Clinical Parameters and Disease Biomarkers in Transcriptomic Datasets." *Biomolecules*, vol. 14, no. 12, 25 Nov. 2024, p. 1501, <https://doi.org/10.3390/biom14121501>.
- Ponce-Bobadilla, Ana Victoria, et al. "Practical Guide to SHAP Analysis: Explaining Supervised Machine Learning Model Predictions in Drug Development." *Clinical and Translational Science*, vol. 17, no. 11, 28 Oct. 2024, <https://doi.org/10.1111/cts.70056>.